Do VLMs See in English?
Mechanistic Analysis of the Language Pivot in Multilingual Vision-Language Models

Anonymous ACL submission

Abstract

Multilingual vision-language models (VLMs) consistently underperform on non-English visual queries, yet the internal mechanism behind this disparity remains unknown. We apply logit-lens analysis and causal activation patching to LLaVA-1.5-7B and show that non-English visual queries are routed through an English-biased representational bottleneck in layers 5–17, extending the English-pivot phenomenon of Wendler et al. (2024) to the multimodal setting. Peak causal influence occurs at layer 8 (ΔAIE=0.49), with the pivot mediated entirely through text-token representations—visual tokens contribute zero independent causal weight. Without visual context, language-specific representations do not emerge at any layer, confirming visual content is necessary to activate the pivot pathway. Building on these findings, we derive training-free language-steering vectors at the mechanistically identified pivot layers, improving Russian VQA by +6.5 pp and Portuguese by +1.0 pp on MMMB without any fine-tuning. Our results suggest the English pivot is a structural property of the LLM backbone that multimodal pre-training does not mitigate.

1 Introduction

Multilingual VLMs are deployed globally, yet a persistent gap separates their English and non-English performance on visual question answering (Fan et al., 2025; Song et al., 2025). Prior work attributes this gap to training-data imbalance or insufficient multilingual instruction tuning, and proposes remedies such as language-aware fine-tuning (Fan et al., 2025) or culturally diverse data (Nyandwi et al., 2025). Yet no prior work has asked the mechanistic question: *where* in the network does English bias arise, and what causal role does the visual modality play?

The text-only interpretability literature provides a starting hypothesis. Wendler et al. (2024) show that LLaMA-family models process multilingual input through three phases: an *input space*, a *concept space* where representations become English-biased, and an *output space* where the target language is reconstructed. Ferrando and Costa-jussà (2024) extend this to circuit-level analysis across languages. Neither work addresses VLMs, leaving open whether the visual modality interacts with or suppresses this pivot.

We fill this gap with the first causal mechanistic analysis of the English pivot in a multilingual VLM. Our contributions are:

- We confirm that LLaVA-1.5-7B routes non-English visual queries through an English-biased concept space in **layers 5–17**, with peak causal responsibility at layer 8 (ΔAIE=0.49; §4.3).
- We show that visual context is *necessary* for any meaningful language-specific representation to emerge, but once present, the English pivot runs entirely through text-token positions—visual-token patching produces zero AIE (§4.3).
- We derive training-free steering vectors at the mechanistically identified pivot layers, yielding **+6.5 pp** on Russian and **+1.0 pp** on Portuguese on the MMMB multiple-choice benchmark, without any fine-tuning (§4.4).

2 Related Work

Mechanistic interpretability of multilingual LLMs. Wendler et al. (2024) establish the three-phase English-pivot model for LLaMA via probing and logit lens. Dumas et al.

(2025) confirm the effect holds for cross-lingual generation. Ferrando and Costa-jussà (2024) show language-agnostic circuits are reused across languages, implying the pivot is structural. Brinkmann et al. (2025) extend causal circuit analysis to larger models, and Resck et al. (2025) connect it to the multilingual curse (Conneau et al., 2020). All of these works study text-only models; we extend causal mechanistic analysis to VLMs.

Multilingual vision-language models. Fan et al. (2025) propose PLAST, a language-aware approach requiring fine-tuning. Song et al. (2025) document non-English VQA failure modes without investigating internal mechanisms. Nyandwi et al. (2025) address cultural diversity at the data level. In contrast, we provide a causal mechanistic account and a training-free fix.

Mechanistic interpretability for VLMs. Golovanevsky et al. (2025) apply circuit analysis to LLaVA to study visual robustness under image corruption—targeting the visual pathway rather than the language-processing pathway we study. Our steering intervention is related to representation engineering (Zou et al., 2023), but derived from mechanistically identified pivot layers rather than contrast pairs.

3 Background

LLaVA-1.5-7B. LLaVA-1.5-7B (Liu et al., 2024) connects a CLIP ViT-L/14 visual encoder to a Vicuna-7B backbone via a two-layer MLP projector. An image is encoded into 576 patch tokens (positions 1–576 in the sequence), followed by the tokenised question. The backbone has 32 Llama-2 decoder layers of hidden size 4096.

Logit lens and PMI. The logit lens (nostalgebraist, 2020) projects an intermediate hidden state $\mathbf{h}_\ell$ through the final layer norm and LM head to obtain token probabilities at any depth. We report *pointwise mutual information*: $\mathrm{PMI}_\ell[\tau] = \log p_\ell[\tau] - \log p_0[\tau]$, where p_0 is the LM-head output at a zero hidden state (the model's unconditional prior), following Geva et al. (2022) to remove vocabulary-frequency bias.

Causal activation patching and AIE. Activation patching (Meng et al., 2022) replaces the clean hidden state at layer ℓ with the corresponding state from a corrupted run, measuring the output change. We define the *Average Indirect Effect*:

$$\mathrm{AIE}(\ell) = \mathbb{E}_e\left[p_\ell^{\mathrm{patch}}[\tau_{\mathrm{en}}] - p^{\mathrm{clean}}[\tau_{\mathrm{en}}]\right], \quad (1)$$

where the clean run uses a non-English question and the corrupted run uses the English equivalent on the same image. τ_{en} is the first token of the model's English response, determined dynamically per example (typically "The", id 450; see Appendix B). Positive AIE at layer ℓ means that layer causally carries English-pivot information.

4 Experiments and Results

4.1 Experimental Setup

Model. We use `llava-hf/llava-1.5-7b-hf` in float16, no quantisation or fine-tuning, greedy decoding.

Probing set. We build a 195-example probing set from COCO val2017 (Lin et al., 2014) across 15 object categories (person, car, bus, train, boat, orange, book, clock, knife, spoon, umbrella, bird, cat, dog, apple). For each example we ask the question in four languages: English (en), Portuguese (pt), German (de), and Russian (ru), using forced-choice prompts listing all 15 category names in the target language. We track the first sub-token of each answer word using contextual-subtraction tokenisation to match generation-time tokenisation (Appendix A).

Evaluation benchmarks. We evaluate steering on MMMB (Sun et al., 2025), a multilingual multiple-choice VQA benchmark covering English, Portuguese, and Russian (200 examples each). German is included in the mechanistic analysis (Figures 1–3) and cross-lingual transfer ablation (§4.5), but MMMB does not provide a German split, so German steering is assessed indirectly via transfer.

Baselines. We compare our steered model (α>0) against unmodified LLaVA-1.5-7B (baseline, α=0) and against steering vectors extracted from fixed layer ranges—early (0–10), mid (11–21), and late (22–31)—to test whether mechanistic localisation is necessary.

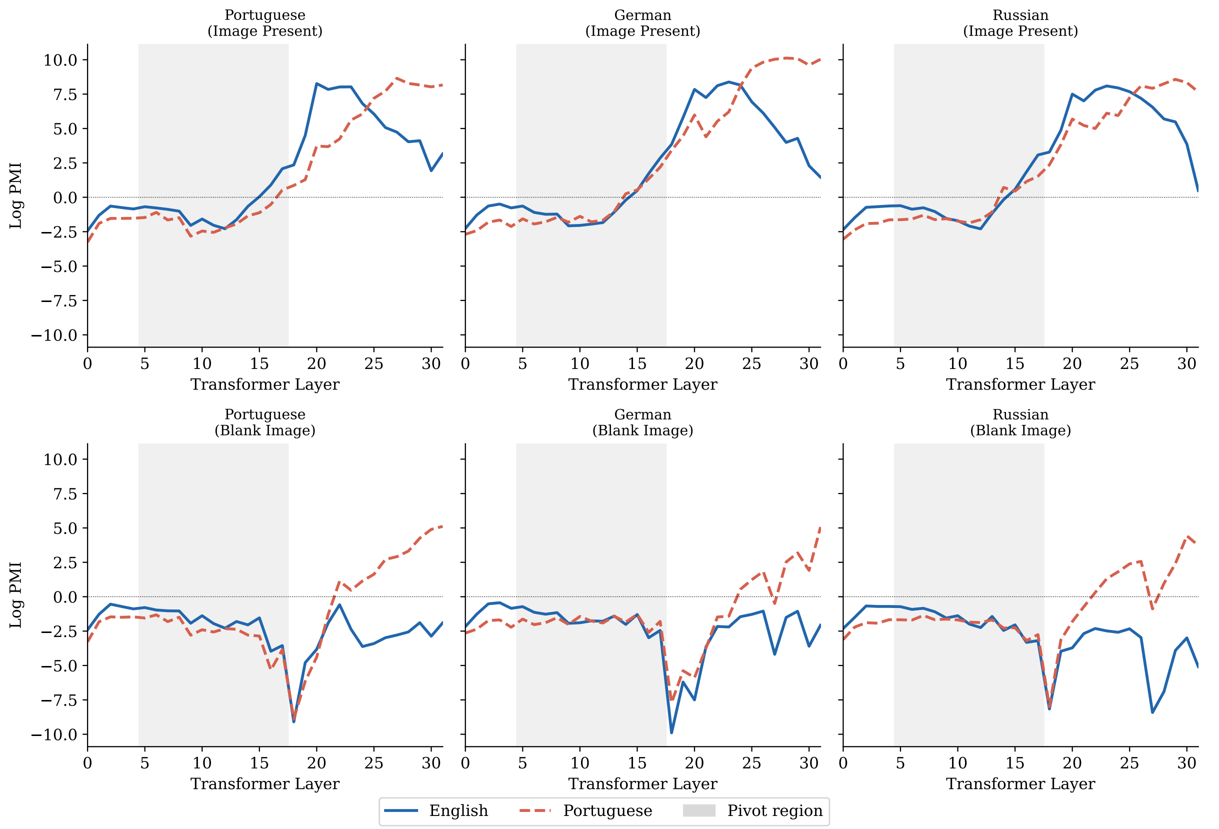

Figure 1: **Logit-lens PMI across layers.** Blue (solid): English answer token; red (dashed): target-language token. Top row: real image; bottom row: blank image. Grey band: mechanistically identified pivot region (layers 5–17). With a real image, both tokens develop strong positive PMI in late layers (20–29), with English peaking first at layers 20–23. Without an image, PMI remains near-zero at every layer—language representations do not emerge without visual context.

4.2 The English Pivot in VLMs

Figure 1 shows logit-lens PMI across all 32 layers.

Image-present condition. With a real image, both English and target-language PMI grow substantially in late layers. English PMI peaks at layers 20–23 (values 8.1–8.4) across all three languages, consistently *before* the target-language peak at layers 27–29 (values 8.6–10.1). This ordering—English representations crystallising 5–7 layers before the target language—directly instantiates Wendler et al.'s concept-space phase in the multimodal setting. The grey pivot region (layers 5–17) corresponds to where English PMI first begins to build, preceding both peaks.

Blank-image condition. Without visual content, PMI values remain near-zero or negative (< -0.4) at every layer for every language. No meaningful language-specific representations form. This confirms that visual input is not merely incidental to the pivot—it is *necessary* to activate the representational pathway that produces language-specific outputs.

4.3 Causal Localisation via Activation Patching

Residual stream patching. Figure 2 (left column) shows strongly positive AIE in layers 5–17 and near-zero thereafter. Peak AIE is at **layer 8** ($\overline{\text{AIE}}$=0.49 averaged across languages), with the signal remaining above the 50%-of-peak threshold (0.24) through layer 17 and dropping to noise floor (< 0.04) by layer 20. We define the **pivot region** as layers 5–17.

The pivot region occupies the first half of the LLM, consistent with Wendler et al.'s concept-space phase, and precedes the late-layer (20–29) representational peaks observed in §4.2.

Attention and MLP patches. Middle and right columns of Figure 2 show near-zero AIE for both attention-output and MLP-output

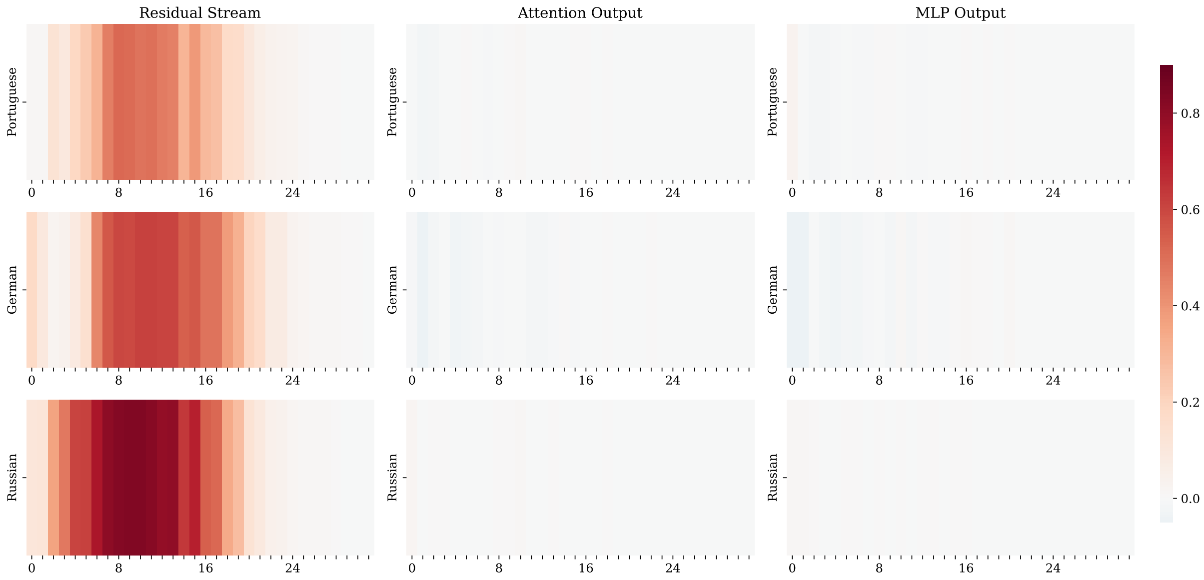

Figure 2: **Average Indirect Effect (AIE) heatmap.** Rows: Portuguese, German, Russian. Columns: residual stream / attention output / MLP output. The residual-stream pivot at layers 5–17 is pronounced across all languages; attention and MLP patches carry negligible weight.

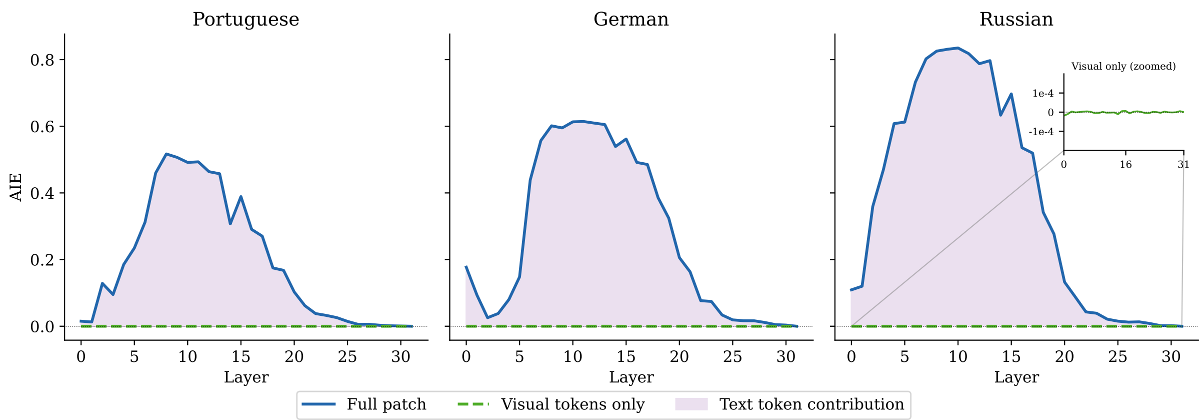

Figure 3: **Full vs. visual-token-only residual patch AIE.** The visual-token curve (green, dashed) is identically zero at every layer: the English pivot is mediated entirely through text-token positions.

patches across all layers (max $|\overline{\mathrm{AIE}}|<0.01$ for attention; <0.01 for MLP except layer 0 where MLP$\approx$0.08, reflecting sub-word tokenisation effects (Geva et al., 2022)). The English pivot operates through the integrated residual stream rather than through separable attention heads or feed-forward networks.

Visual token contribution. Figure 3: the visual-token-only AIE is **zero at every layer across all languages**. This is structurally guaranteed: the same image is used in both the non-English (clean) and English (corrupted) passes, so the 576 CLIP visual token activations are identical and contribute no differential signal. The implication is clear: *the causal pathway of the English pivot runs entirely through text-token positions.* Visual context shapes *when* language-specific representations emerge (as shown by the blank-image condition in §4.2), but does not itself carry English-biasing causal information.

4.4 Training-Free Steering at Pivot Layers

Steering vector extraction. For each non-English language ℓ, we extract a steering vector at each pivot layer $L \in \{5, \ldots, 17\}$:

$$\mathbf{v}_L^{(\ell)} = \frac{1}{N} \sum_{i=1}^{N} \mathbf{h}_L^{(\ell,i)} - \mathbf{h}_L^{(\mathrm{en},i)}, \qquad (2)$$

Table 1: **Main results on MMMB** (4-way multiple-choice accuracy, %). "Ours": best accuracy over $\alpha \in \{0.01, 0.02, 0.05, 0.07, 0.1, 0.2, 0.3, 0.4, 0.5\}$; optimum is α=0.2 for both PT and RU. EN: baseline only (no steering vector for English). Δ: gain over baseline.

Method	EN	PT	RU
Baseline	57.5	63.0	51.5
Ours (α=0.2)	57.5	**64.0**	**58.0**
Δ	—	+1.0	+6.5

Table 2: **Layer range ablation on MMMB (best α per cell, %).** Best α shown in parentheses where it differs from 0.2.

Layer range	PT	RU
Early (0–10)	63.5 (α=0.2)	51.5 (α=0.01)
Mid (11–21)	63.0 (α=0.01)	54.5 (α=0.1)
Late (22–31)	63.0 (α=0.01)	51.5 (α=0.01)
Ours (5–17)	**64.0** (α=0.2)	**58.0** (α=0.2)

where $\mathbf{h}_L^{(\cdot,i)}$ is the residual-stream hidden state at the last answer position for example i (N=195). Split-half cosine similarity exceeds 0.85 across all languages and pivot layers, indicating stable steering directions.

Steered inference. At inference time we add $\alpha\,\mathbf{v}_L^{(\ell)}$ to the hidden state at each pivot layer via an in-place forward hook, applied to the last active position at every decode step (KV-cache compatible; Appendix B).

Results (Table 1). Steering yields **+6.5 pp** on Russian MMMB (51.5% → 58.0% at α=0.2), nearly closing the 6-point gap between Russian and English baselines. Portuguese gains +1.0 pp (63.0% → 64.0%); the smaller absolute gain reflects the already high Portuguese baseline, which exceeds English—likely due to MMMB test-set difficulty variation across languages.

4.5 Ablations

Layer range ablation (Table 2). Our mechanistic pivot region (5–17) outperforms all fixed ranges on both languages. The early range is next best for Portuguese (+0.5 pp over baseline), consistent with the AIE peak occurring in the early-to-mid network. Mid and late ranges match or fall below baseline for Russian, indicating that steering later layers is ineffective. The gap between ours and the best fixed range is +6.5 vs. +3.0 pp (mid) for Russian, confirming that mechanistic localisation is necessary.

Table 3: **Visual token ablation on MMMB (%).** "Steer" = best α: α=0.2 (real image), α=0.02 (blank image).

Condition	PT Base	PT Steer	RU Base	RU Steer
Real image	63.0	64.0	51.5	58.0
Blank image	42.0	41.5	46.5	48.5

Table 4: **Cross-lingual transfer on MMMB (best α per cell, %).** "—" = own-language result (see Table 1).

Vector source	MMMB-PT	MMMB-RU
PT (own)	—	55.5 (α=0.3)
DE	64.0 (α=0.2)	58.0 (α=0.2)
RU (own)	63.5 (α=0.3)	—

Visual token ablation (Table 3). Blank images drastically lower baseline performance (PT: −21 pp; RU: −5 pp), confirming visual context is necessary for correct answering. With blank images, steering is marginal or slightly harmful, consistent with the logit-lens finding that visual input is required for strong language-specific representations to form.

Steering coefficient sensitivity (Figure 4). Performance peaks at α=0.2 for both MMMB languages and degrades sharply beyond α=0.3. Large α values corrupt generation quality (MMMB-RU drops to 0% at α=0.5, indicating degenerate outputs).

Cross-lingual transfer (Table 4). The **German steering vector transfers perfectly** to both Portuguese (64.0% = own-language) and Russian (58.0% = own-language). In contrast, the Portuguese vector applied to Russian yields 55.5% (−2.5 pp vs. own-language). The full transferability of the German vector suggests it captures a general "non-English" axis rather than a language-specific direction, consistent with the shared concept space observed in activation patching.

5 Discussion

Why does the pivot exist in VLMs? Our results are consistent with the view that LLaVA-1.5-7B inherits the English pivot from its Vicuna backbone. The LLM backbone is pretrained predominantly on English; non-English

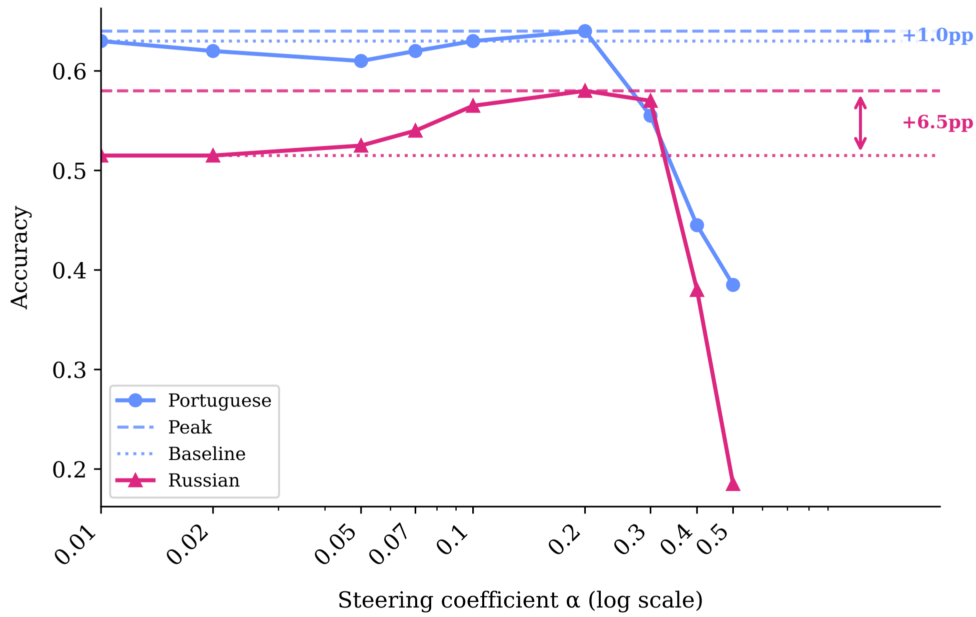

Figure 4: **Sensitivity to steering coefficient α on MMMB.** Performance peaks at α=0.2 and degrades sharply beyond α=0.3. The effective window ($\alpha \in [0.05, 0.2]$) is consistent with prior representation-engineering results (Zou et al., 2023).

sequences are mapped into a concept space built around English representations. Multimodal pre-training adds a visual input modality but does not restructure this internal language topology. Visual tokens activate the language-processing pathway but do not themselves carry a language-direction signal—explaining why visual-token patching yields zero AIE.

Steering direction geometry. The complete transferability of the German vector (Table 4) suggests that all three steering directions (pt, de, ru) approximately coincide—pointing along a single "non-English language" axis rather than three language-specific directions. This is consistent with the shared concept-space hypothesis of Wendler et al. (2024) and raises the possibility that a single universal vector could suffice for multilingual correction.

Limitations. This work studies a single VLM (LLaVA-1.5-7B) on three non-English languages; generalisation to other architectures and language families requires further investigation. Our steering requires a 195-example probing set per language, assuming access to target-language COCO annotations. The zero visual-token AIE follows from using the same image in both passes; using paired images with language-relevant visual differences would probe whether visual content can carry pivot information in more ecologically valid settings.

6 Conclusion

LLaVA-1.5-7B processes non-English visual queries through an English-biased representational bottleneck in layers 5–17, with peak causal influence at layer 8 (ΔAIE=0.49). Visual context is necessary for language-specific representations to form at all, but once present, the pivot is mediated entirely through text-token positions. Training-free steering vectors derived from these mechanistically identified layers improve Russian MMMB by +6.5 pp and Portuguese MMMB by +1.0 pp, with German vectors transferring perfectly across languages. These findings establish the English pivot as a structural property of the LLM backbone and provide a concrete, mechanistically grounded target for future multilingual VLM design.

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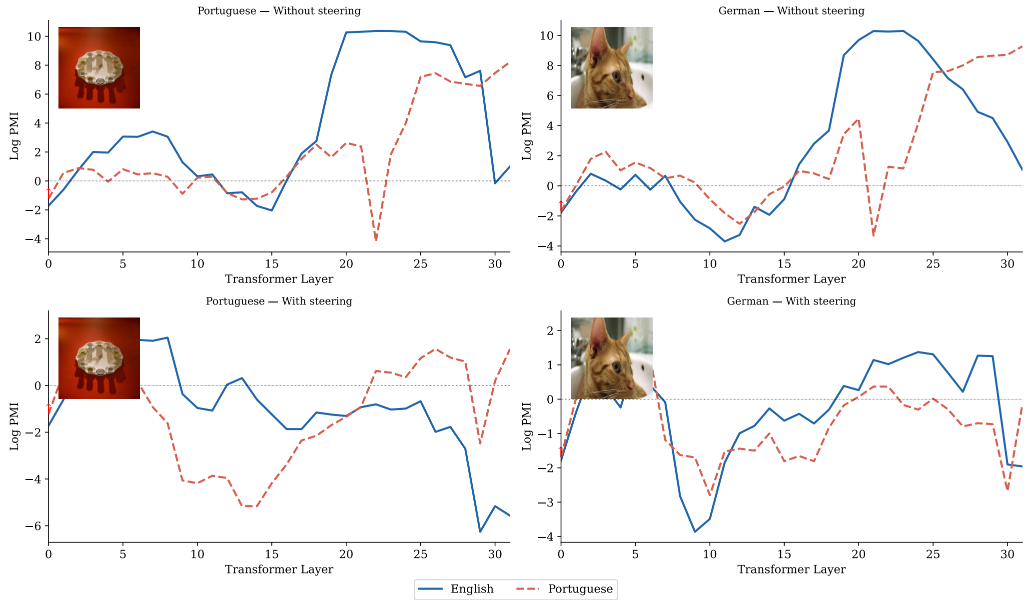

Figure 5: **Qualitative logit-lens walkthrough.** Two examples where the English pivot is most severe (Portuguese, left; German, right), shown without steering (top) and with steering at α=0.2 (bottom). Red arrow: layer where target-language PMI first exceeds English PMI. Inset: COCO image. Steering shifts the crossover to an earlier layer, consistent with target-language representations being amplified at the pivot layers.

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A Probing Set Details

Table 5 lists the 15 COCO categories and their Portuguese and German answer tokens. All answer words are tracked by their first sub-token under the Vicuna tokeniser, using contextual subtraction (encoding "ASSISTANT: {word}" and stripping the prefix tokens) to match generation-time tokenisation. The leading-space token (id 29871) is filtered before selecting the first sub-token.

Table 5: Object categories with PT and DE answer tokens. Multi-token words tracked by first sub-token only (†).

Category	PT	DE
person	pessoa	Person
car	carro	Auto
bus	ônibus†	Bus
train	trem	Zug
boat	barco	Boot
orange	laranja†	Orange
book	livro	Buch
clock	relógio†	Uhr
knife	faca	Messer†
spoon	colher†	Löffel†
umbrella	guarda-chuva†	Regenschirm†
bird	pássaro†	Vogel†
cat	gato	Katze†
dog	cachorro†	Hund
apple	maçã†	Apfel†

B Implementation Details

Sequence alignment. Non-English questions are typically 2–5 tokens longer than English counterparts. We keep the clean (non-English) input at full length to preserve the correct `answer_pos`, and truncate only the corrupted (English) input to `min_len`.

Pre-hook patching under transformers 5.0. Transformers 5.0 wraps each `LlamaDecoderLayer` in a `GradientCheckpointingLayer` that discards post-hook return values. We register patching as a *pre-hook* on layer ℓ+1 (which is outside the gradient-checkpointing wrapper), rather than a post-hook on layer ℓ. For the final layer (ℓ=31), the pre-hook is on `model.language_model.model.norm`.

Response-token target. The model's first generated token is consistently the article "The" (id 450) rather than the object name, as the model answers in sentence form. We determine τ_{en} dynamically per example via a one-token greedy generation on the English input. Steering is applied in-place at the last active token position at each generate step: $\mathbf{h}_L \mathrel{+}= \alpha \cdot \mathbf{v}_L^{(\ell)}$ via a forward hook on `model.language_model.model.layers[L]`. This correctly steers both the prefill position and each autoregressive decode step under KV-cache.